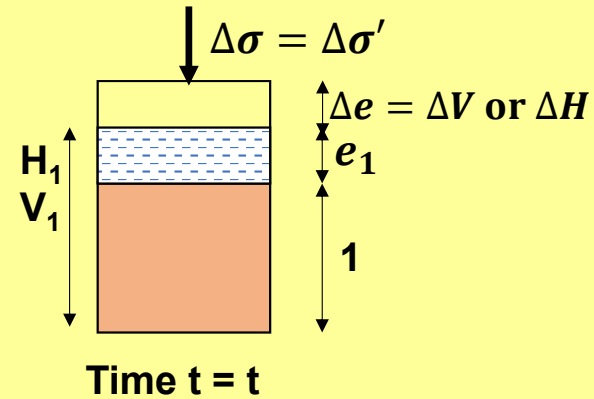
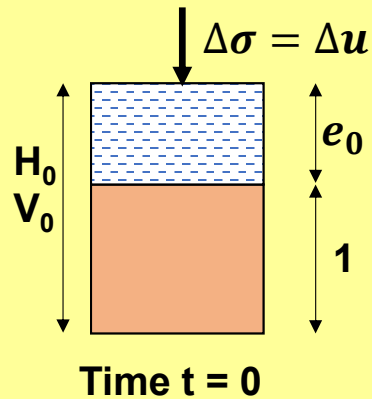


Module 8
Lecture 8.1b Soil consolidation: Terminologies

Soil consolidation

Consider saturated soil mass



External load induce $\Delta\sigma$ and increases excess pore water pressure Δu

With passage of time, Δu dissipates and finally $\Delta\sigma = \Delta\sigma'$

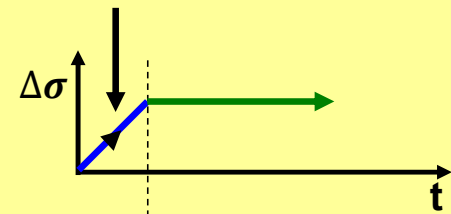
This causes volume change $\Delta e = \Delta V$ or ΔH

For a 1-D case, $\Delta e = \Delta V = \Delta H$

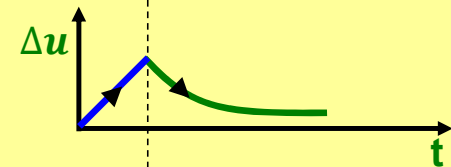
Lateral strain is negligible or restricted

Stress change with time during consolidation

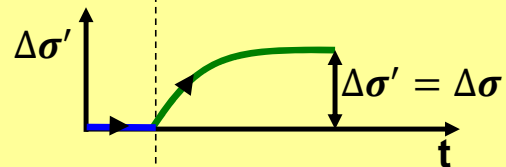
Time to construction



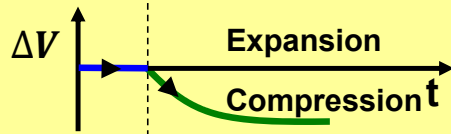
Beyond time of construction external pressure is constant



After construction phase, u dissipates



Loss of u is gain in σ'



Gain in σ' results in volume change/ settlement as a function of time

Rate of dissipation of u and gain in σ' essentially depends on k of soil

1D consolidation: Some important terminologies

$$\text{Volumetric strain} = \frac{\Delta V}{V} = \frac{\Delta H}{H} = \frac{\Delta e}{1 + e_0}$$

ΔH represents settlement caused due to loading

$$\Delta H = \frac{\Delta e}{1 + e_0} H_0 \quad H_0 : \text{initial height of soil layer}$$

Coefficient of volume compressibility (m_v)

$$m_v = \frac{\Delta e}{1 + e_0} \frac{1}{\Delta \sigma'}$$

This is modulus of volume change

Defined as the change in volume per unit volume per unit increase in effective stress

Unit is m^2/kN

Consolidation settlement (S_c)

$$S_c = \Delta H = \frac{\Delta e}{1 + e_0} H_0$$

$$S_c = \frac{\Delta e}{1 + e_0} H_0 \frac{\Delta \sigma'}{\Delta \sigma'}$$

$$S_c = m_v H_0 \Delta \sigma'$$

Coefficient of compressibility (a_v)

$$a_v = \frac{\Delta e}{\Delta \sigma'}$$

$$m_v = \frac{a_v}{1 + e}$$